A numerical algorithm for finding pinch points within Briggs' method

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It has long been known that the symmetries of the linearized Navier-Stokes equations form the basis for their linear stability theory. In particular, the translation in time and space plus the scaling symmetry of the dependent variables implies the Fourier-Laplace transform, and thus the closely related the normal-mode approach. Moreover, for certain base flows such as linear shear, further symmetries exist that induce further solutions such as the Kelvin mode or other families of modes. Here we consider two further symmetries, i.e. the Galilean invariance and the rotation invariance, and its implications for the dispersion relation $\mathscr{D} = 0$. From the first, it follows that the dispersion relation on the complex eigenvalues ω must be analytic or holomorphic. Singular eigenvalues are thus physically excluded, similarly to the constraints on the contour of the integration path in the complex ω-plane induced by the causality condition. The above result is also independent of whether the instabilities are purely temporal or spatiotemporal, as used in Briggs theory, for example. Rotational invariance implies that ... We conclude by giving an example in which we show that ...

I. INTRODUCTION

A famous result of linear stability theory is Squire's theorem. Squire showed by comparing temporal normal modes in 2D and in 3D that the 2D modes correspond to a lower Reynolds number than their corresponding 3D modes.[1] Therefore, 2D modes are critical for the onset o instability.

However, Squire's theorem only applies to temporal normal modes. Oberlack et al. extended the idea of comparing modes between 2D and 3D to spatiotemporal normal modes.[2] Interestingly, the opposite of Squire's theorem may hold in that case: there may be 3D perturbations with a lower Reynolds number than their corresponding 2D perturbations. Wilhelm et al. then applied the extended Squire transformation on the physical initial value problem and demonstrated how the temporal asymptotic of the 3D initial value problem can be reduced to a 2D initial value problem.[3]

However, the investigation of individual of these spatio-temporal normal modes is meaningless, since the spatial part of their eigenfunction diverges, even initially. Nevertheless, they can still be used inside a framework like Briggs' method in a superposition of these spatio-temporal modes.

Briggs' method provides a framework for determining the temporal or spatial asymptotics of linear initial-value problems by analyzing the analytic structure of the dispersion relation in the complex planes of wave frequency and wave number. Rather than focusing on individual perturbation modes, the method tracks how singularities of the dispersion relation interact under analytic continuation and contour deformation. The key outcome is the identification of pinch points and saddle points that govern the asymptotic response. In this sense, Briggs' method assigns physical meaning to spatiotemporal modes through their collective contribution to the asymptotic behavior of the flow.[4]

There exist various techniques to determine the location of such pinch points, or of saddle points in general, in the complex α-plane. A naive approach would be to map out the dispersion relation $\mathscr{D}(\alpha, \omega)$ entirely.

Deissler assumed the position of a saddle point for plane Poiseuille flow to be initially located on the real wave number axis, at the point of maximal temporal growth for a real wave number, and tracked the position of this saddle point for a range of Reynolds numbers over varying observer velocities.[5] He did not track the saddle points to pinch points, i.e. to zero observer velocity, as the lower edge rays of neutral growth still showed positive observer velocities, meaning that they correspond to convective instabilities. He supplemented this with an asymptotic analysis for large Reynolds numbers, and concluded that the instabilities of plane Poiseuille flow are always of convective nature.

Sengupta and Orkwis applied the Newton method using numerical Jacobians to find pinch points in the complex α-plane.[6] By doing so, and by using the property of analyticity of the dispersion relation, they were able to develop a method that only uses three eigenvalue computations per Newton step.

The disadvantage of these methods is that even if a pinch point is found, it is not guaranteed that the spatial integration contour can actually be continuously deformed to pass through this pinch point. Furthermore, it is not clear a priori whether the pinching branches originate from different half-planes, which is a necessary condition for the pinching.[7] If both branches originate from the same half-plane, the contour can always be continuously deformed to avoid the singularity.

II. GOVERNING EQUATIONS

We analyze an incompressible Newtonian fluid governed by the non-dimensionalized Navier-Stokes equations

$$\boldsymbol{\nabla}\cdot\boldsymbol{u}=0, \tag{1a}$$

$$\frac{\partial \boldsymbol{u}}{\partial t}+(\boldsymbol{u}\cdot\boldsymbol{\nabla})\boldsymbol{u}=-\boldsymbol{\nabla}p+\frac{1}{Re}\nabla^2\boldsymbol{u},\ \text{with } Re:=\frac{\rho U_0 L}{\mu}, \tag{1b}$$

where $\boldsymbol{u}$, p, and Re respectively denote velocity, pressure and the Reynolds number. The Reynolds number comprises the density ρ, a characteristic velocity U_0, a characteristic length scale L, and the dynamic viscosity μ.

Let a Cartesian coordinate system with coordinates $\boldsymbol{x}=(x,y,z)$ for the streamwise, wall-normal and spanwise directions be introduced with the corresponding velocities $\boldsymbol{u}=(u,v,w)$. For the further analysis of linear stability theory, we perform the splitting of our flow field variables $\{\boldsymbol{u},p\}$ into a laminar flow field $\{\boldsymbol{U}=(U,V,W),P\}$ and a superimposed perturbation flow field $\{\boldsymbol{u}'=(u',v',w'),p'\}$, i.e.

$$\{\boldsymbol{u},p\}=\{\boldsymbol{U},P\}+\varepsilon\{\boldsymbol{u}',p'\}\quad\text{with}\quad\varepsilon\ll 1. \tag{2}$$

Further, we assume to have a parallel shear flow, i.e.

$$\boldsymbol{U}=(U(y),0,0)\,. \tag{3}$$

We have the boundary conditions on the boundary $\partial\Omega$ and the initial condition

$$\{U|_{\partial\Omega_y}=U_b,\,\partial_x P=P_x\},\quad \{\boldsymbol{u}',p'\}|_{\partial\Omega}=0, \tag{4a}$$

$$\{\boldsymbol{u}',p'\}|_{t=0}=\{\boldsymbol{u}'_0,p'_0\}. \tag{4b}$$

After solving the laminar flow field $\{U,P\}$ up to an arbitrary additive constant for the pressure, with the above assumptions, using (1a) and eliminating the pressure perturbation p', equation (1b) linearizes into two equations for the perturbation quantities $\{v',\eta'\}$, i.e.

$$\left[\left(\frac{\partial}{\partial t}+U\frac{\partial}{\partial x}\right)\nabla^2-U''\frac{\partial}{\partial x}-\frac{1}{Re}\nabla^4\right]v'=0, \tag{5a}$$

$$\left[\frac{\partial}{\partial t}+U\frac{\partial}{\partial x}-\frac{1}{Re}\nabla^2\right]\eta'=-U'\frac{\partial v'}{\partial z}, \tag{5b}$$

where v' is the wall-normal velocity perturbation, and η' is the wall-normal vorticity perturbation, defined as

$$\eta'=\frac{\partial u'}{\partial z}-\frac{\partial w'}{\partial x}. \tag{6}$$

After solving v' in (5a), η' can be solved via the equation for the vorticity perturbation (5b), and p' can be calculated via the y-momentum equation in (1b) up to an arbitrary additive constant. After solving η', $\{u',w'\}$ can be calculated via the continuity equation (1a) and the definition of η' (6).

Allow the functions $\{v',\eta'\}$ to be reformulated with the normal mode approach into amplitude functions $\{\tilde{v},\tilde{\eta}\}$, dependent on the inhomogeneous y-direction, and an exponential with a wave frequency ω in time t and wave numbers α and β in the homogeneous x- and z-directions, respectively

$$\{v',\eta'\}(x,y,z,t)=\{\tilde{v},\tilde{\eta}\}(y)e^{i(\alpha x+\beta z-\omega t)}. \tag{7}$$

This is related to a Fourier transform in space and a Laplace transform in time. The normal mode approach corresponds to selecting one single Fourier-Laplace mode, whereas the full Fourier-Laplace transform represents a superposition of such Fourier-Laplace modes.

With (7), the spatiotemporal PDEs (5) are transformed into the ODEs for $\{\tilde{v},\tilde{\eta}\}$

$$\left[(-i\omega+i\alpha U)\left(\frac{\mathrm{d}^2}{\mathrm{d}y^2}-\left(\alpha^2+\beta^2\right)\right)-i\alpha U''-\frac{1}{Re}\left(\frac{\mathrm{d}^2}{\mathrm{d}y^2}-\left(\alpha^2+\beta^2\right)\right)^2\right]\tilde{v}=0, \tag{8a}$$

$$\left[(-i\omega+i\alpha U)-\frac{1}{Re}\left(\frac{\mathrm{d}^2}{\mathrm{d}y^2}-\left(\alpha^2+\beta^2\right)\right)\right]\tilde{\eta}=-i\beta U'\tilde{v}, \tag{8b}$$

the first of which is the Orr-Sommerfeld equation (OSE) and the second is the Squire equation (SE).[8,9] In general, we have $\omega,\alpha,\beta\in\mathbb{C}$.

Furthermore, we will analyze the 2D problem by setting $\beta=0$. In this limit, η' becomes identically zero, and the Squire equation (5b) decouples and becomes trivial. The perturbation dynamics is then governed solely by the OSE. That being so, u' is then fully determined by v' via the continuity equation (1a).

III. BRIGGS' METHOD

1. The physical initial value problem

We consider the 2D physical initial value problem for one fixed Reynolds number Re, with the inverse Fourier-Laplace transform of the wall-normal velocity perturbation v' as

$$v(x,y,t)=\frac{1}{4\pi^2}\int_{\mathscr{L}_\omega}\int_{\mathscr{F}_\alpha}\tilde{v}(\alpha,y,\omega)e^{i(\alpha x-\omega t)}\,\mathrm{d}\alpha\,\mathrm{d}\omega, \tag{9}$$

where $\mathscr{L}_\omega$ is the temporal integration contour in the complex ω-plane, and $\mathscr{F}_\alpha$ is the spatial integration contour in the complex α-plane. The integration contours are initially parametrized as

$$\mathscr{L}_\omega:\ s\mapsto\omega(s)=s+i\sigma,\quad s\in(-\infty,\infty),\ \sigma\in\mathbb{R} \tag{10a}$$

$$\mathscr{F}_\alpha:\ s\mapsto\alpha(s)=s, \tag{10b}$$

where σ is chosen such that the temporal integration contour $\mathscr{L}_\omega$ is a straight line with constant imaginary part above all singularities of the Green's kernel G. These are the roots of the dispersion relation $\mathscr{D}$, since $G\propto 1/\mathscr{D}$. The spatial integration contour $\mathscr{F}_\alpha$ is initially along the real axis.

The temporal integration contour $\mathscr{L}_\omega$ can be closed with a clockwise semicircle below for $t>0$ using Jordan's lemma. The spatial integration contour $\mathscr{F}_\alpha$ can be closed with a counterclockwise semicircle above for $x>0$, and with a clockwise semicircle below for $x<0$, respectively using Jordan's lemma.

The spectral eigenfunction $\tilde{v}$ is given as

$$\tilde{v}(\alpha,y,\omega) = \int_{\Omega_y} G(\alpha,y,\omega;y')\tilde{v}_0(\alpha,y')\,\mathrm{d}y', \tag{11a}$$

$$\tilde{v}_0(\alpha,y) = \int_{-\infty}^{\infty} v_0'(x,y)e^{-i\alpha x}\,\mathrm{d}x, \tag{11b}$$

where G is the Green's kernel of the differential operator in spectral space, i.e. the spectral solution to a Dirac impulse in the y-direction $\delta(y-y')$, $\tilde{v}_0$ is the spectral initial condition, i.e. the Fourier-Laplace transform of the physical initial condition v_0' and Ω_y is the y-domain.

A forcing term $f'(x,y,t)$ on the right-hand side of the equation for the wall-normal velocity perturbation v', (5a), yields the following the spectral solution

$$\tilde{v}(\alpha,y,\omega) = \int_{\Omega_y} G(\alpha,y,\omega;y')\tilde{v}_0(\alpha,y')\,\mathrm{d}y' + \int_{\Omega_y} G(\alpha,y,\omega;y')\tilde{f}(\alpha,y',\omega)\,\mathrm{d}y', \tag{12a}$$

$$\tilde{f}(\alpha,y,\omega) = \int_0^{\infty}\int_{-\infty}^{\infty} f'(x,y,t)e^{-i(\alpha x-\omega t)}\,\mathrm{d}x\,\mathrm{d}t. \tag{12b}$$

We set the initial condition $v_0' = \tilde{v}_0 = 0$ and work with the forcing f' and $\tilde{f}$ respectively instead. We now examine two canonical forcing scenarios commonly used in Briggs-type analyses,

A. Dirac impulse forcing in space and time, representing the fundamental Green's function response, and

B. Time-periodic forcing, corresponding to a harmonic input at a prescribed temporal frequency ω_f.

A. Dirac impulse forcing in space and time

In this first scenario, the forcing is

$$f'(x,y,t) = \delta(x)\delta(y-y')\delta(t) \tag{13a}$$

$$\tilde{f}(y) = \delta(y-y'). \tag{13b}$$

We want to analyze the temporal asymptotic of (9) for $t\to\infty$ to determine the long-time behavior of the perturbation responding to the initial excitation (13).

Each point of an integration contour corresponds to a whole eigenvalue spectrum in the other complex plane. For example, a wave number α on the spatial integration contour $\mathscr{F}_\alpha$ produces a whole temporal eigenvalue spectrum. Of all the temporal eigenvalues in that spectrum, the most unstable eigenvalue is dominant, i.e. the one with the largest imaginary part. Now going over all α in $\mathscr{F}_\alpha$, a temporal eigenvalue trajectory $\omega(\mathscr{F}_\alpha)$ forms. The same is true for the temporal integration contour, which produces for each wave frequency ω on $\mathscr{L}_\omega$ a whole spectrum of spatial eigenvalues. This time, two spatial eigenvalues from each spectrum are tracked, namely those closest to the spatial integration contour $\mathscr{F}_\alpha$ above and below, as they govern the topology of the integration contour. Going over all ω in $\mathscr{L}_\omega$, two spatial eigenvalue trajectories form, $\alpha_u(\mathscr{L}_\omega)$ above and $\alpha_l(\mathscr{L}_\omega)$ below $\mathscr{F}_\alpha$.

Using the tools of complex analysis, one can continuously deform the integration contours without changing the value of the inverse Fourier-Laplace transform, as long as the topology of the eigenvalue trajectories in the complex planes stays intact. This means that eigenvalues initially above or below the integration contours respectively stay above or below. At a pinch point α_p, where the eigenvalue trajectories $\alpha_u(\mathscr{L}_\omega)$ and $\alpha_l(\mathscr{L}_\omega)$ coalesce, the spatial integration contour can no longer be deformed without breaking the topology. This pinch point corresponds to a double root of the dispersion relation, since the eigenvalues are nothing but roots of the dispersion relation $\mathscr{D}$ and two roots coalesce here, so

$$\mathscr{D}(\alpha,\omega;Re)|_{(\alpha_p,\omega_p)} = 0, \qquad \left.\frac{\partial\mathscr{D}(\alpha,\omega;Re)}{\partial\alpha}\right|_{(\alpha_p,\omega_p)} = 0. \tag{14}$$

When integrating the inverse Fourier-Laplace transform considering $\omega(\alpha)$, the phase of the kernel exponent is stationary with respect to α along the ray x/t

$$\frac{\partial(\alpha x-\omega t)}{\partial\alpha} = x - \frac{\mathrm{d}\omega}{\mathrm{d}\alpha}t \overset{!}{=} 0, \tag{15a}$$

$$\frac{\mathrm{d}\omega}{\mathrm{d}\alpha} = \frac{x}{t}, \tag{15b}$$

and one can see by implicit differentiation,

$$\left.\frac{\mathrm{d}\omega}{\mathrm{d}\alpha}\right|_{(\alpha_p,\omega_p)} = -\left.\frac{\frac{\partial\mathscr{D}(\alpha,\omega;Re)}{\partial\alpha}}{\frac{\partial\mathscr{D}(\alpha,\omega;Re)}{\partial\omega}}\right|_{(\alpha_p,\omega_p)} = 0, \tag{16}$$

that this group velocity is zero at the double root $(\partial\mathscr{D}(\alpha,\omega;Re)/\partial\alpha)|_{(\alpha_p,\omega_p)} = 0$. This mean that the pinch point (α_p,ω_p) constitutes a saddle point in the stationary reference frame corresponding to the ray $x/t = 0$. Therefore, the spatial inverse transform can be evaluated at α_p using the method of steepest descent, since the integration contour $\mathscr{F}_\alpha$ passes through that saddle point in the direction of steepest descent.

The position of the corresponding ω_p in the complex ω-plane determines the nature of the instability: if $\omega_{p,i} > 0$, the disturbance is absolutely unstable, meaning that it grows for $t\to\infty$ at any position x. If $\omega_{p,i} < 0$, the disturbance is stable in that frame of reference, meaning that the disturbance decays for $t\to\infty$ at any position x. The temporal inverse transform can be evaluated by closing the integration contour $\mathscr{L}_\omega$ using the residuum theorem at ω_p.

It is important to note that Briggs' method only detects absolute instabilities in the stationary reference frame corresponding to $x/t = 0$. Absolute instabilities in a moving reference frame, i.e. convective instabilities in the stationary reference frame, cannot be captured directly.

In general, the observer can move from the origin with velocity $V_G \in \mathbb{R}$ along a ray $x/t = V_G$ as

$$x = V_G t. \tag{17}$$

Substituting this into the exponent leads to

$$e^{i(\alpha x-\omega t)} = e^{i(\alpha V_G-\omega)t}. \tag{18}$$

Along that ray, one observes only temporal growth. One can see that the phase of the kernel exponent is stationary with respect to α at

$$\frac{\partial(\alpha V_G t-\omega t)}{\partial\alpha} = V_G t - \frac{\mathrm{d}\omega}{\mathrm{d}\alpha} t \overset{!}{=} 0, \tag{19a}$$

$$\frac{\mathrm{d}\omega}{\mathrm{d}\alpha} = V_G. \tag{19b}$$

Each ray $x/t = V_G$ has a different saddle point $(\alpha_s(V_G), \omega_s(V_G))$ that is in general different from the pinch point (α_p, ω_p) found by the coalescing $\alpha_u(\mathscr{L})$ and $\alpha_l(\mathscr{L})$. Only for $V_G = 0$, both the pinch point and the saddle point coincide. Along rays with

$$\omega_{s,i} - \alpha_{s,i} V_G > 0, \tag{20}$$

one observes temporal growth. If they exist, the regions of rays with temporal growth are bounded by two observer velocities $V_{G,l}$, $V_{G,u}$, with $V_{G,l} < V_{G,u}$, for which exactly

$$\omega_{s,i} - \alpha_{s,i} V_G = 0. \tag{21}$$

These velocities $V_{G,l}$ and $V_{G,u}$ are the spreading speeds of the disturbance wave packet. Growth takes place only for rays between with $V_{G,l} < V_G < V_{G,u}$.

If $V_{G,l} < 0 < V_{G,u}$, the instability constitutes an absolute instability, or in other words, if (20) holds for $V_G = 0$. Evidently, (20) then reduces to $\omega_{p,i} > 0$, as previously established.

However, this is now a mechanism to classify convective instabilities. If $0 \leq V_{G,l}$ or $V_{G,u} \leq 0$, the instability constitutes a convective instability.

B. Time-periodic forcing

Here, the forcing is

$$f'(x,y,t) = \delta(x)\delta(y-y')e^{-i\omega_f t} \tag{22a}$$

$$\tilde{f}(y,\omega) = \frac{\delta(y-y')}{\omega-\omega_f}, \tag{22b}$$

Notice that this forcing (22) already fulfills causality by definition, since the Laplace transform is defined over $t \in [0,\infty)$.

In this second scenario, we want to analyze the spatial asymptotic of (9) for $x \to \infty$ to determine the far-field behavior of the perturbation responding to a continuous time-periodic forcing at frequency ω_f.

For the time-periodic forcing, the temporal integration contour has to go around the excitation frequency ω_f on the real axis in a Hankel-type manner. For $\sigma < 0$, the integration contour runs horizontally with imaginary part σ until $\omega_f - \varepsilon$, $\varepsilon \ll 1$, then vertically up to the real axis, then a semicircle with radius ε around the excitation frequency ω_f, then back vertically and horizontally as before.

The temporal integration contour $\mathscr{L}_\omega$ is parametrized piecewise as the union

$$\mathscr{L}_\omega = \bigcup_{j=1}^{5} \Gamma_j, \tag{23}$$

where the individual segments are given by

$$\Gamma_1: \quad m \mapsto \omega(m) = m + i\sigma, \qquad m \in \left(-\infty,\ \omega_f - \varepsilon\right], \tag{24a}$$

$$\Gamma_2: \quad m \mapsto \omega(m) = \omega_f - \varepsilon + im, \quad m \in [\sigma,\ 0], \tag{24b}$$

$$\Gamma_3: \quad \varphi \mapsto \omega(\varphi) = \omega_f + \varepsilon e^{i\varphi}, \qquad \varphi \in [\pi,\ 0], \tag{24c}$$

$$\Gamma_4: \quad m \mapsto \omega(m) = \omega_f + \varepsilon + im, \quad m \in [0,\ \sigma], \tag{24d}$$

$$\Gamma_5: \quad m \mapsto \omega(m) = m + i\sigma, \qquad m \in \left[\omega_f + \varepsilon,\ \infty\right). \tag{24e}$$

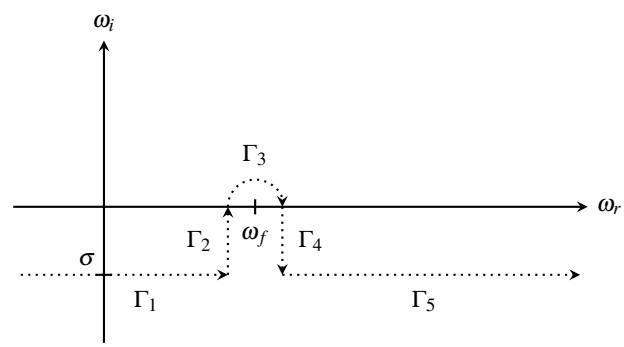

FIG. 1. Piecewise parametrized integration contour $\mathscr{L}_\omega$.

Figure 1 shows how the integration contour $\mathscr{L}_\omega$ is parametrized piecewise in the complex ω-plane for $\sigma < 0$.

For the evaluation of (9), the temporal inverse transform is evaluated with the residuum theorem at ω_f, yielding the remaining spatial inverse transform as

$$v(x,y,t) = \frac{1}{4\pi^2}\int_{\mathscr{L}_\omega}\int_{\mathscr{F}_\alpha} \frac{G(\alpha,y,\omega;y')}{\omega-\omega_f} e^{i(\alpha x-\omega t)}\,\mathrm{d}\alpha\,\mathrm{d}\omega, \tag{25a}$$

$$= e^{-i\omega_f t}\frac{i}{2\pi}\int_{\mathscr{F}_\alpha} G\left(\alpha,y,\omega_f;y'\right) e^{i\alpha x}\,\mathrm{d}\alpha. \tag{25b}$$

The remaining spatial inverse transform in (25) is now - analogously to the Dirac impulse forcing case - continuously deformed, as long as the topology of the eigenvalue trajectories stays intact. This deformation is ultimately limited by the coalescence of two spatial eigenvalues according to (14) with $\omega = \omega_f$, and (25) is evaluated asymptotically for $x \to \infty$ using the method of steepest descent at that pinch point.

IV. NUMERICAL ALGORITHM

A. Evolution equations of the integration contours

The key idea is to set up evolution equations for the integration contours, defining a geometric gradient-flow problem

in the complex planes. Therein, the integration contours are treated as elastic curves repelled by the eigenvalue trajectory (trajectories) of the respective other integration contour.

The integration contours are continuously deformed over a pseudo-time τ, i.e. an optimization time that relaxes the integration contours into an "optimal" configuration.

We parametrize the integration contours as functions in the complex plane, where we assign an imaginary part to every real part of the contour. For the temporal integration contour $\mathscr{L}_\omega$, all real parts have the same imaginary part, so this contour is and stays a straight horizontal line in the complex ω-plane. For the spatial integration contour $\mathscr{F}_\alpha$, each real part can have a different imaginary part,

$$\mathscr{L}_\omega : \mathbb{R} \to \mathbb{C}, \quad \omega_r \mapsto \omega = \omega_r + i\omega_i, \tag{26a}$$
$$\mathscr{F}_\alpha : \mathbb{R} \to \mathbb{C}, \quad \alpha_r \mapsto \alpha = \alpha_r + i\alpha_i(\alpha_r). \tag{26b}$$

For the time-periodic forcing case, only the horizontal parts of the integration contour are defined with this function relation, and the evolution equations are only valid for those parts. As $\varepsilon \ll 1$, this should not pose a problem.

We set up the following evolution equations

$$\partial_\tau \mathscr{L}_\omega = -\boldsymbol{n}\left(\nabla_{\boldsymbol{n}} \Phi_{\mathscr{L}_\omega} + \sigma_{\text{curv}}\kappa\right) - i\lambda + \boldsymbol{t}\cdot\boldsymbol{v}_\parallel, \tag{27a}$$
$$\partial_\tau \mathscr{F}_\alpha = -\boldsymbol{n}\left(\nabla_{\boldsymbol{n}} \Phi_{\mathscr{F}_\alpha} + \sigma_{\text{curv}}\kappa\right) + \boldsymbol{t}\cdot\boldsymbol{v}_\parallel. \tag{27b}$$

Herein, we measure the distance between a point of the integration contour and the eigenvalue trajectory (trajectories) of the respective other integration contour with a potential function Φ. With the directional normal derivative $-\nabla_{\boldsymbol{n}}$, the integration contours move so as to maximize their distance from the eigenvalue trajectory (trajectories). Only normal motion alters the geometry of a contour, while tangential motion would merely change its parametrization. Therefore, all physically relevant forces are projected onto the normal direction with the unit normal vector $\boldsymbol{n}$.

The second term involves the curvature κ and a surface tension analogue σ_{curv}. This curvature regularization term penalizes high curvatures in order to prevent cusps and high-frequency oscillations, stabilizing the evolution and enforcing smooth contours.

For the temporal integration contour, $-i\lambda$ enforces a global downward shift in the complex ω-plane, acting as a Lagrange multiplier fixing the imaginary level of the integration contour. It executes the main idea of Briggs' method to lower the temporal integration contour as much as possible.

The tangential velocity $\boldsymbol{v}_\parallel$, projected onto the tangential direction with the unit tangential vector $\boldsymbol{t}$, is a control term that vanishes in the final form of the evolution equations.

For $f \in \{\omega_i, \alpha_i(\alpha_r)\}$, the geometric quantities can be obtained as

$$\boldsymbol{n} = \frac{-f' + i}{\sqrt{1 + (f')^2}}, \tag{28a}$$
$$\boldsymbol{t} = \frac{1 + if'}{\sqrt{1 + (f')^2}}, \tag{28b}$$
$$\kappa = \boldsymbol{\nabla}\cdot\boldsymbol{n} = \frac{-f''}{\left(1 + (f')^2\right)^{\frac{3}{2}}}, \tag{28c}$$
$$\nabla_{\boldsymbol{n}} = \boldsymbol{n}\cdot\boldsymbol{\nabla}. \tag{28d}$$

With that, the evolution equations can be rewritten as

$$\partial_\tau \omega_i = -\frac{\partial \Phi_{\mathscr{L}_\omega}}{\partial \omega_i} - \lambda, \tag{29a}$$
$$\partial_\tau \alpha_i = \left(\frac{\mathrm{d}\alpha_i}{\mathrm{d}\alpha_r}\frac{\partial \Phi_{\mathscr{F}_\alpha}}{\partial \alpha_r} - \frac{\partial \Phi_{\mathscr{F}_\alpha}}{\partial \alpha_i}\right) + \frac{\sigma_{\text{curv}}\frac{\mathrm{d}^2\alpha_i}{\mathrm{d}\alpha_r^2}}{1 + \left(\frac{\mathrm{d}\alpha_i}{\mathrm{d}\alpha_r}\right)^2}. \tag{29b}$$

Notice that derivative terms of $\mathrm{d}\omega_i/\mathrm{d}\omega_r$ and $\mathrm{d}^2\omega_i/\mathrm{d}\omega_r^2$ have canceled, since the temporal integration contour is and stays a straight horizontal line.

For the global potential functions, we choose

$$\Phi_{\mathscr{L}_\omega}(\omega) = \int_{\omega(\mathscr{F}_\alpha)} \varphi\left(\left|\omega - \omega'\right|\right)\left|\mathrm{d}\omega'\right|, \tag{30a}$$
$$\Phi_{\mathscr{F}_\alpha}(\alpha) = \sum_{j\in\{u,l\}} \int_{\alpha_j(\mathscr{L}_\omega)} \varphi\left(\left|\alpha - \alpha'\right|\right)\left|\mathrm{d}\alpha'\right|. \tag{30b}$$

These functions measure for every point ω, α on the integration contours $\mathscr{L}_\omega$, $\mathscr{F}_\alpha$ the distance to all points ω', α' on the eigenvalue trajectories $\omega(\mathscr{F}_\alpha)$, $\alpha_u(\mathscr{L}_\omega)$, $\alpha_l(\mathscr{L}_\omega)$, by integrating a local potential function φ over the eigenvalue trajectories.

For the local potential function, we choose a function that is smooth and analytic, and decays exponentially at long distance, such as

$$\varphi(r) = e^{\frac{\zeta}{r^2+\varepsilon}} - 1. \tag{31}$$

Herein, the parameter $\zeta \in \mathbb{R}^+$ controls the effective width of the potential, where larger values lead to a steeper and more localized repulsion of the integration contour from the eigenvalue trajectories, while smaller values result in a broader and weaker interaction. The regularization parameter ε prevents the potential from becoming singular when the integration contour locally touches an eigenvalue trajectory, guaranteeing analyticity of φ (and its derivatives) for all r, in particular also for $r = 0$.

This local potential function has the analytic derivative

$$\frac{\mathrm{d}\varphi(r)}{\mathrm{d}r} = -\frac{2\zeta r}{(r^2+\varepsilon)^2}\, e^{\frac{\zeta}{r^2+\varepsilon}}, \tag{32}$$

so that the derivatives of the global potential functions are

$$\frac{\partial \Phi_{\mathscr{L}_\omega}(\omega)}{\partial \omega_i} = \int_{\omega(\mathscr{F}_\alpha)} \frac{\omega_i - \omega_i'}{|\omega - \omega'|^2} \frac{\mathrm{d}\varphi\left(|\omega - \omega'|\right)}{\mathrm{d}\,|\omega - \omega'|} \left|\mathrm{d}\omega'\right|, \quad (33a)$$

$$\frac{\partial \Phi_{\mathscr{F}_\alpha}(\alpha)}{\partial \alpha_r} = \sum_{j\in\{u,l\}} \int_{\alpha_j(\mathscr{L}_\omega)} \frac{\alpha_r - \alpha_r'}{|\alpha - \alpha'|^2} \frac{\mathrm{d}\varphi\left(|\alpha - \alpha'|\right)}{\mathrm{d}\,|\alpha - \alpha'|} \left|\mathrm{d}\alpha'\right|, \quad (33b)$$

$$\frac{\partial \Phi_{\mathscr{F}_\alpha}(\alpha)}{\partial \alpha_i} = \sum_{j\in\{u,l\}} \int_{\alpha_j(\mathscr{L}_\omega)} \frac{\alpha_i - \alpha_i'}{|\alpha - \alpha'|^2} \frac{\mathrm{d}\varphi\left(|\alpha - \alpha'|\right)}{\mathrm{d}\,|\alpha - \alpha'|} \left|\mathrm{d}\alpha'\right|. \quad (33c)$$

B. Discrete symmetries

In order to restrict the region in the complex planes that need to be investigated, we use the discrete symmetries of the OSE (8a) in the form $\mathscr{L}\tilde{v} = 0$. These discrete symmetries are not to be confused with continuous symmetries, as e.g. used in Lie symmetry theory (see e.g. Wilhelm et al.[3]).

The first discrete symmetry is complex conjugation. Complex conjugation $\overline{(\cdot)}$ yields $\overline{\mathscr{L}}\bar{\tilde{v}} = 0$. The complex conjugate of the differential operator $\overline{\mathscr{L}}$ only affects the complex spectral parameters (α, ω) to $(\overline{\alpha}, \overline{\omega})$. Like so, it follows that the eigenvalues of $\mathscr{L}$ and $\overline{\mathscr{L}}$ are related as

$$\mathscr{D}(\alpha, \omega) = \mathscr{D}(\overline{\alpha}, \overline{\omega}) = 0. \quad (34)$$

The second discrete symmetry is reversal of the normal mode phase in space and time, which follows from the normal mode approach (7). A complex conjugate normal mode corresponds to the physical field

$$e^{\bar{i}(\overline{\alpha}x - \overline{\omega}t)} = e^{i[(-\overline{\alpha})x - (-\overline{\omega})t]}, \quad (35)$$

which is itself again a normal mode, but with spectral parameters $(-\overline{\alpha}, -\overline{\omega})$. Therefore, the dispersion relation is also symmetric to

$$\mathscr{D}(\alpha, \omega) = \mathscr{D}(-\overline{\alpha}, -\overline{\omega}) = 0. \quad (36)$$

Combining both symmetries gives

$$\mathscr{D}(\alpha, \omega) = \mathscr{D}(-\alpha, -\omega) = \mathscr{D}(\overline{\alpha}, \overline{\omega}) = \mathscr{D}(-\overline{\alpha}, -\overline{\omega}) = 0. \quad (37)$$

Consequently, it is sufficient to restrict the investigation to a representative region in the complex planes. Note that one can only restrict the region for one spectral parameter. We choose $\alpha_r \geq 0$, so the full ω_r has to be investigated. The remaining region in the complex planes can then be recovered by applying the symmetry relations (37).

C. Numerical implementation

The integration contours are discretized by sampling the real axis at a finite number of points N_ω, N_α. We introduce a uniform discretization in the intervals $[\omega_{r,\min}, \omega_{r,\max}]$ and $[0, \alpha_{r,\max}]$, where $(\cdot)_{(j)}$ denotes a discretized point on an integration contour in the complex plane.

The spatial derivatives are approximated with central finite difference quotients as

$$\frac{\mathrm{d}\alpha_{i,(j)}}{\mathrm{d}\alpha_r} = \frac{\alpha_{i,(j+1)} - \alpha_{i,(j-1)}}{\alpha_{r,(j+1)} - \alpha_{r,(j-1)}}, \quad (38a)$$

$$\frac{\mathrm{d}^2\alpha_{i,(j)}}{\mathrm{d}\alpha_r^2} = \frac{\alpha_{i,(j+1)} - 2\alpha_{i,(j)} + \alpha_{i,(j-1)}}{\left(\alpha_{r,(j+1)} - \alpha_{r,(j)}\right)\left(\alpha_{r,(j)} - \alpha_{r,(j-1)}\right)}, \quad (38b)$$

where the end points respectively use forward or backward finite difference quotients. The pseudo-time derivative is approximated with a forward finite difference quotient, i.e. explicit, with pseudo-time step length Δ_τ, where $(\cdot)^{(n)}$ denotes an discretized point in pseudo-time, as

$$\partial_\tau \omega_{i,(j)}^{(n)} = \frac{\omega_{i,(j)}^{(n+1)} - \omega_{i,(j)}^{(n)}}{\Delta\tau}, \quad (39a)$$

$$\partial_\tau \alpha_{i,(j)}^{(n)} = \frac{\alpha_{i,(j)}^{(n+1)} - \alpha_{i,(j)}^{(n)}}{\Delta\tau}. \quad (39b)$$

The pseudo-time evolution is continued until convergence, defined by

$$\max_j \left|\alpha_{i,(j)}^{(n+1)} - \alpha_{i,(j)}^{(n)}\right| < \delta, \quad (40)$$

with a prescribed tolerance δ. At convergence, the integration contours are stationary solutions of the geometric evolution equations and represent admissible Briggs integration contours for the subsequent asymptotic analysis.

V. VALIDATION WITH ANALYTIC TEST EQUATION

The algorithm is applied to the analytic test equation[6]

$$\mathscr{D}(\alpha, \omega) = \omega - i\left(\frac{\alpha^3}{3} + \alpha^2 + 3\alpha\right) = 0 \quad (41)$$

with the

VI. COMPARISON WITH OTHER ALGORITHMS

VII. CONCLUSION AND OUTLOOK

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